

EX NO:1	Butterworth IIR Filters
DATE:	1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

CODE :-

```

clc;

clear;

close all

% Filter Specifications

Fs = 10000;    % Sampling Frequency (Hz)

Fp = 1000;    % Passband Frequency (Hz)

Fst = 1500;    % Stopband Frequency (Hz)

Rp = 1;        % Passband Ripple (dB)

Rs = 60;        % Stopband Attenuation (dB)

% Determine Filter Order

[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth Low-Pass Filter

[b, a] = butter(N, Wn, 'low');

% Frequency Response

figure;

freqz(b, a);

title('Magnitude and Phase Response');

% Impulse Response

figure;

[h, n] = impz(b, a, 50);

stem(n, h, 'filled');

grid on;

title('Impulse Response');

xlabel('Samples');

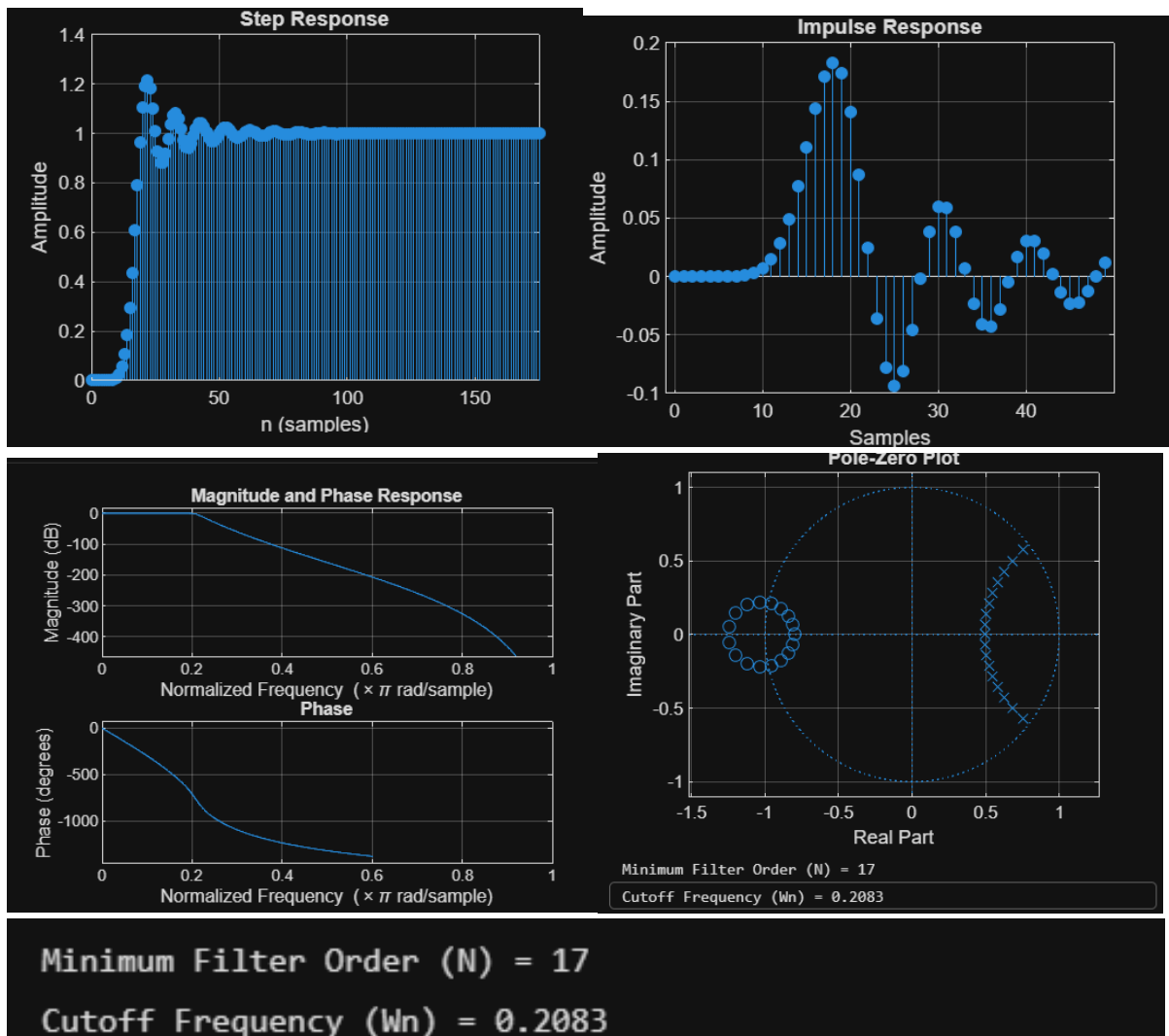
ylabel('Amplitude');
```

```
% Step Response
figure;
stepz(b, a);
grid on;
title('Step Response');

% Pole-Zero Plot
figure;
zplane(b, a);
grid on;
title('Pole-Zero Plot');

% Display Filter Parameters
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

OUTPUT :-



EX NO:1	Butterworth IIR Filters
DATE:	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

CODE :-

```

clc;

clear;

close all;

% Filter Specifications

Fs = 10000;    % Sampling Frequency (Hz)

Fp = 3000;    % Passband Frequency (Hz)

Fst = 2000;    % Stopband Frequency (Hz)

Rp = 1;        % Passband Ripple (dB)

Rs = 60;        % Stopband Attenuation (dB)

% Determine Filter Order

[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth High-Pass Filter

[b, a] = butter(N, Wn, 'high');

% Frequency Response

figure;

freqz(b, a);

title('Magnitude and Phase Response');

% Impulse Response

figure;

[h, n] = impz(b, a, 50);

stem(n, h, 'filled');

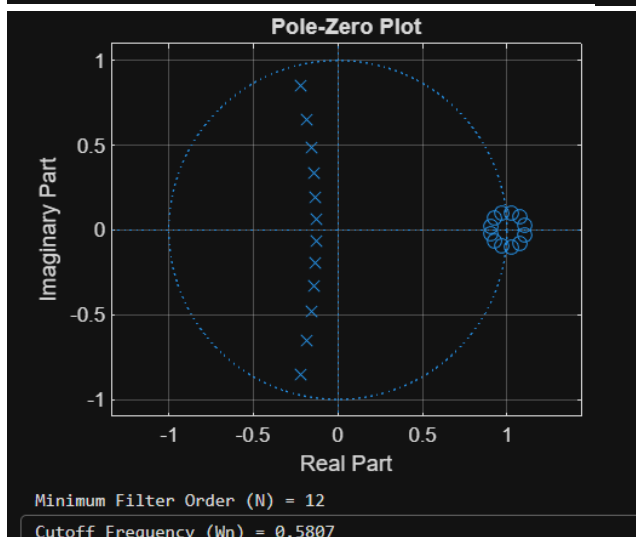
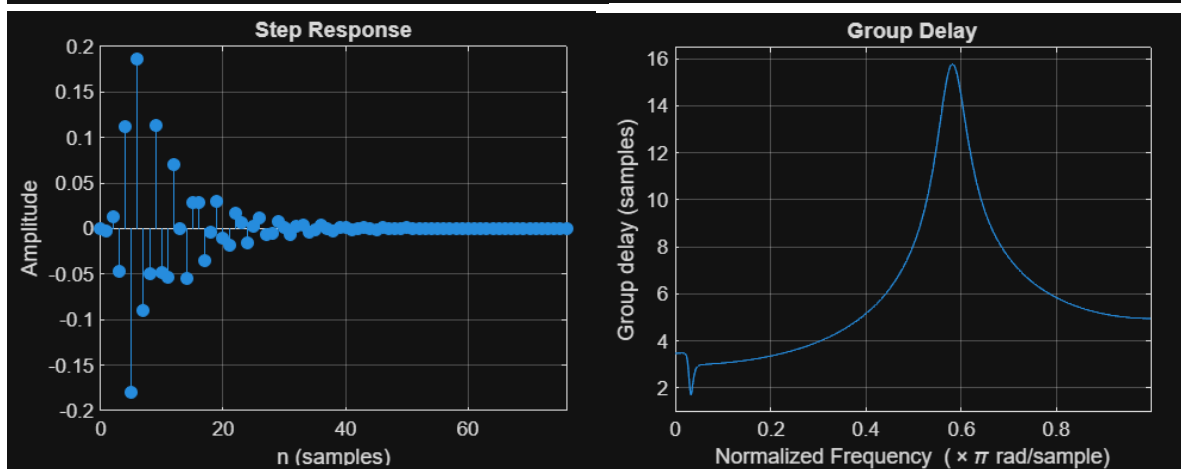
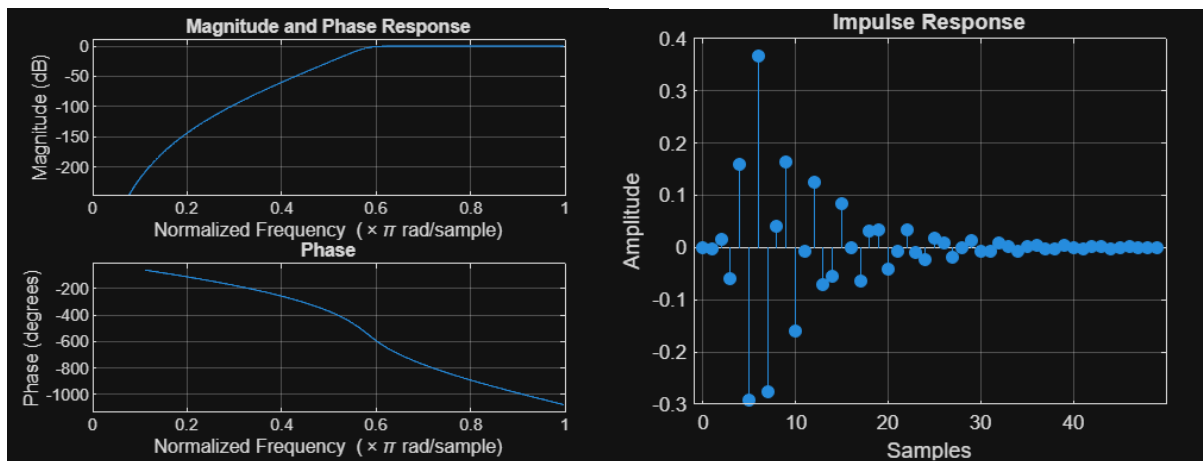
grid on;

title('Impulse Response');

xlabel('Samples');
```

```
ylabel('Amplitude');  
  
% Step Response  
  
figure;  
stepz(b, a);  
grid on;  
title('Step Response');  
  
% Group Delay  
  
figure;  
grpdelay(b, a);  
grid on;  
title('Group Delay');  
  
% Pole-Zero Plot  
  
figure;  
zplane(b, a);  
grid on;  
title('Pole-Zero Plot');  
  
% Display Filter Parameters  
  
fprintf('Minimum Filter Order (N) = %d\n', N);  
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

OUTPUT :-



Minimum Filter Order (N) = 12
Cutoff Frequency (ω_n) = 0.5807